R2 Newton's second law

Name:

Class:

Date:

Time: **364 minutes**

Marks: **304 marks**

Comments:

Q1.

An object of mass 5 kg is moving in a straight line.

As a result of experiencing a forward force of F newtons and a resistant force of R newtons it accelerates at 0.6 m s^{-2}

Which one of the following equations is correct?

Circle your answer.

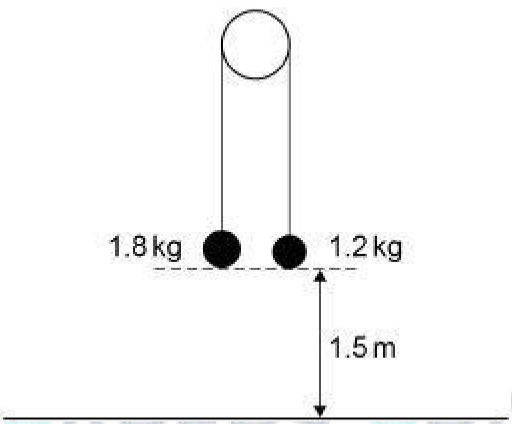
$F - R = 0$ $\quad$ $F - R = 5$ $\quad$ $F - R = 3$ $\quad$ $F - R = 0.6$

(Total 1 mark)

Q2.

In this question use g = 9.8 m s^{-2}

Two particles, of mass 1.8 kg and 1.2 kg, are connected by a light, inextensible string over a smooth peg.



(a) Initially the particles are held at rest 1.5 m above horizontal ground and the string between them is taut.

The particles are released from rest.

Find the time taken for the 1.8 kg particle to reach the ground.

(5)

(b) State one assumption you have made in answering part (a).

(1)

(Total 6 marks)

Q3.

A cyclist, Laura, is travelling in a straight line on a horizontal road at a constant speed of 25 km h^{-1}

A second cyclist, Jason, is riding closely and directly behind Laura. He is also moving with a constant speed of 25 km h^{-1}

(a) The driving force applied by Jason is likely to be less than the driving force applied by Laura.

Explain why.

(1)

(b) Jason has a problem and stops, but Laura continues at the same constant speed.

Laura sees an accident 40 m ahead, so she stops pedalling and applies the brakes.

She experiences a total resistance force of 40 N

Laura and her cycle have a combined mass of 64 kg

(i) Determine whether Laura stops before reaching the accident.

Fully justify your answer.

(4)

(ii) State one assumption you have made that could affect your answer to part (b)(i).

(1)

(Total 6 marks)

Q4.

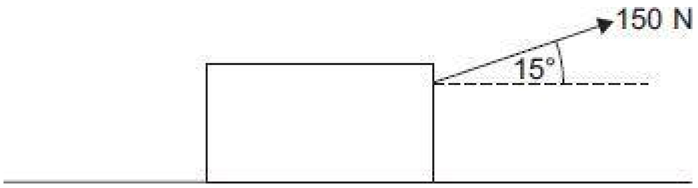
In this question use g = 9.8 m s^{-2}

A boy attempts to move a wooden crate of mass 20 kg along horizontal ground. The coefficient of friction between the crate and the ground is 0.85

(a) The boy applies a horizontal force of 150 N. Show that the crate remains stationary.

(3)

(b) Instead, the boy uses a handle to pull the crate forward. He exerts a force of 150 N, at an angle of 15° above the horizontal, as shown in the diagram.



Determine whether the crate remains stationary.

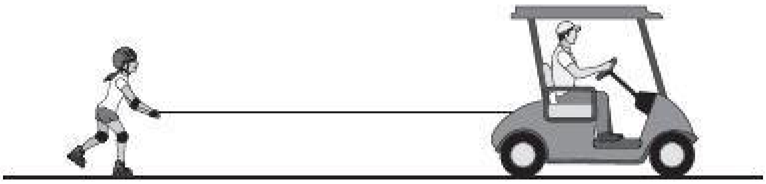
Fully justify your answer.

(5)

(Total 8 marks)

Q5.

A buggy is pulling a roller-skater, in a straight line along a horizontal road, by means of a connecting rope as shown in the diagram.



The combined mass of the buggy and driver is 410 kg
A driving force of 300 N and a total resistance force of 140 N act on the buggy.

The mass of the roller-skater is 72 kg
A total resistance force of *R* newtons acts on the roller-skater.

The buggy and the roller-skater have an acceleration of 0.2 m s^{-2}

(a) (i) Find *R*.

(3)

(ii) Find the tension in the rope.

(3)

(b) State a necessary assumption that you have made.

(1)

(c) The roller-skater releases the rope at a point *A*, when she reaches a speed of 6 m s^{-1}

She continues to move forward, experiencing the same resistance force.

The driver notices a change in motion of the buggy, and brings it to rest at a distance of 20 m from *A*.

(i) Determine whether the roller-skater will stop before reaching the stationary buggy.

Fully justify your answer.

(5)

(ii) Explain the change in motion that the driver noticed.

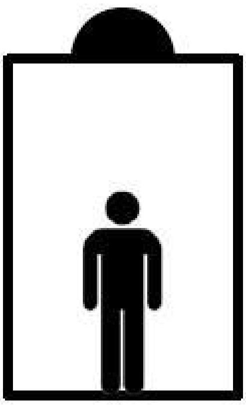
(2)

(Total 14 marks)

Q6.

In this question use g = 10 m s^{-2} , giving your final answers to an appropriate degree of accuracy.

A man of mass 80 kg is travelling in a lift.
The lift is rising vertically.



The lift decelerates at a rate of 1.5 m s^{-2}

Find the magnitude of the force exerted on the man by the lift.

(Total 3 marks)

Q7.

A particle, of mass 400 grams, is initially at rest at the point O.

The particle starts to move in a straight line so that its velocity, v m s^{-1}, at time t seconds is given by

$$v = 6t^2 - 12t^3 \text{ for } t > 0$$

(a) Find an expression, in terms of t, for the force acting on the particle.

(3)

(b) Find the time when the particle next passes through O.

(5)

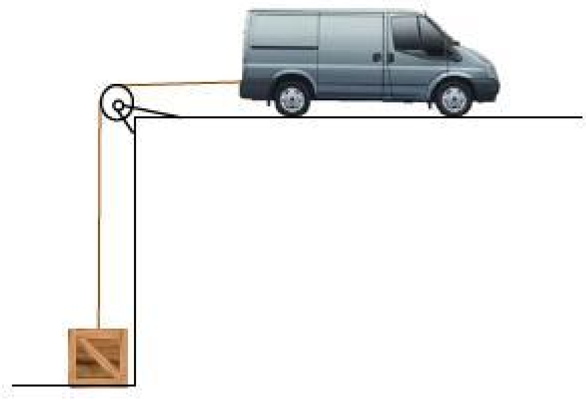
(Total 8 marks)

Q8.

In this question use g = 9.8 m s^{-2}, giving your final answers to an appropriate degree of accuracy.

A van of mass 1300 kg and a crate of mass 300 kg are connected by a light inextensible rope.
The rope passes over a light smooth pulley, as shown in the diagram.

The rope between the pulley and the van is horizontal.



Initially, the van is at rest and the crate rests on the lower level. The rope is taut.
The van moves away from the pulley to lift the crate from the lower level.
The van's engine produces a constant driving force of 5000 N.
A constant resistance force of magnitude 780 N acts on the van.
Assume there is no resistance force acting on the crate.

(a) (i) Draw a diagram to show the forces acting on the crate while it is being lifted.

(1)

(ii) Draw a diagram to show the forces acting on the van while the crate is being lifted.

(1)

(b) Show that the acceleration of the van is 0.80 m s^{-2}

(4)

(c) Find the tension in the rope.

(2)

(d) Suggest how the assumption of a constant resistance force could be refined to produce a better model.

(1)

(Total 9 marks)

Q9.

A single force of magnitude 4 newtons acts on a particle of mass 50 grams.

Find the magnitude of the acceleration of the particle.

Circle your answer.

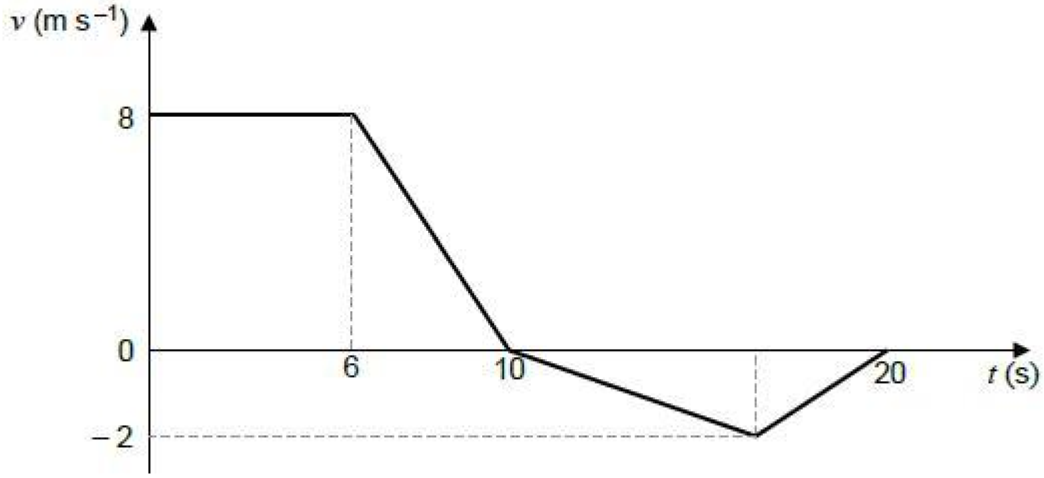
12.5 m s^{-2} 0.08 m s^{-2} 0.0125 m s^{-2} 80 m s^{-2}

(Total 1 mark)

Q10.

The graph below models the velocity of a small train as it moves on a straight track for 20 seconds.

The front of the train is at the point A when $t = 0$
The mass of the train is 800kg.



(a) Find the total distance travelled in the 20 seconds.

(3)

(b) Find the distance of the front of the train from the point A at the end of the 20 seconds.

(1)

(c) Find the maximum magnitude of the resultant force acting on the train.

(2)

(d) Explain why, in reality, the graph may not be an accurate model of the motion of the train.

(1)
(Total 7 marks)

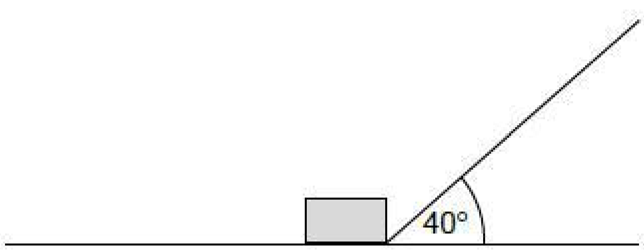
Q11.

In this question use g = 9.8 m s^{-2}, giving your final answers to an appropriate degree of accuracy.

The diagram shows a box, of mass 8.0 kg, being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.

The tension in the string is 50 newtons.



The coefficient of friction between the box and the board is μ

Model the box as a particle.

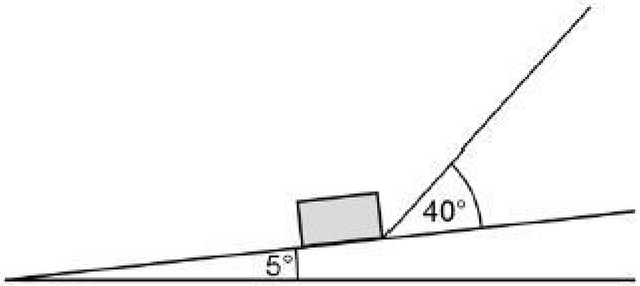
(a) Show that $\mu = 0.83$

(4)

(b) One end of the board is lifted up so that the board is now inclined at an angle of 5° to the horizontal.

The box is pulled up the inclined board.
The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at 3 m s^{-2}



(i) Draw a diagram to show the forces acting on the box as it moves.

(1)

(ii) Find the tension in the string as the box accelerates up the slope at 3 m s^{-2}.

(7)

(Total 12 marks)

Q12.

Three forces act on a particle. These forces are $(9\mathbf{i} - 3\mathbf{j})$ newtons, $(5\mathbf{i}\ \ 8\mathbf{j})$ newtons and $(-7\mathbf{i} + 3\mathbf{j})$ newtons. The vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

(a) Find the resultant of these forces.

(2)

(b) Find the magnitude of the resultant force.

(2)

(c) Given that the particle has mass 5 kg, find the magnitude of the acceleration of the particle.

(2)

(d) Find the angle between the resultant force and the unit vector $\mathbf{i}$.

(3)

(Total 9 marks)

Q13.

A box, of mass 3 kg, is placed on a rough slope inclined at an angle of 40° to the horizontal. It is released from rest and slides down the slope.

(a) Draw a diagram to show the forces acting on the box.

(1)

(b) Find the magnitude of the normal reaction force acting on the box.

(2)

(c) The coefficient of friction between the box and the slope is 0.2. Find the magnitude of the friction force acting on the box.

(2)

(d) Find the acceleration of the box.

(3)

(e) State an assumption that you have made about the forces acting on the box.

(1)

(Total 9 marks)

Q14.

A tractor, of mass 3500 kg, is used to tow a trailer, of mass 2400 kg, across a horizontal field. The trailer is connected to the tractor by a horizontal tow bar. As they move, a constant resistance force of 800 newtons acts on the trailer and a constant resistance force of R newtons acts on the tractor. A forward driving force of 2500 newtons acts on the tractor. The trailer and tractor accelerate at 0.2 m s^{-2}.

(a) Find R.

(3)

(b) Find the magnitude of the force that the tow bar exerts on the trailer.

(3)

(c) State the magnitude of the force that the tow bar exerts on the tractor.

(1)

(Total 7 marks)

Q15.

A particle, of mass 12 kg, is moving along a straight horizontal line. At time t seconds, the particle has speed v m s^{-1}. As the particle moves, it experiences a resistance force of magnitude $4v^{\frac{1}{3}}$. No other horizontal force acts on the particle.

The initial speed of the particle is 8 m s^{-1}.

(a) Show that

$$v = \left(4 - \frac{2}{9}t\right)^{\frac{3}{2}}$$

(6)

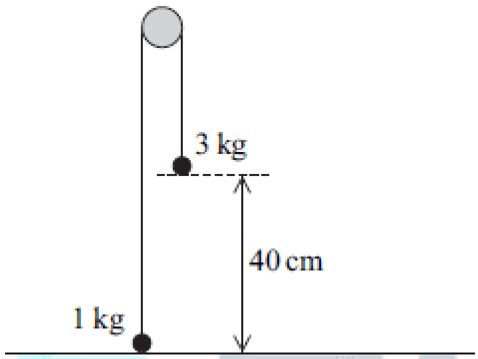
(b) Find the value of t when the particle comes to rest.

(1)

(Total 7 marks)

Q16.

Two particles are connected by a light inextensible string that passes over a smooth peg. The particles have masses of 3 kg and 1 kg. The 1 kg particle is pulled down to ground level, where it is 40 cm below the level of the 3 kg particle, as shown in the diagram.



The particles are released from rest with the string vertical above each particle. Assume that no resistance forces act on the particles as they move.

(a) By forming two equations of motion, one for each particle, find the magnitude of the acceleration of the particles after they have been released but before the 3 kg particle hits the ground.

(5)

(b) Find the speed of the 1 kg particle when the 3 kg particle hits the ground.

(2)

(c) After the 3 kg particle has hit the ground, the 1 kg particle continues to move and the string is now slack. Find the maximum height above ground level reached by the 1 kg particle.

(3)

(d) If a constant air resistance force also acts on the particles as they move, explain how this would change your answer for the acceleration in part (a). Give a reason for your answer.

(2)

(Total 12 marks)

Q17.

A block of mass 30 kg is dragged across a rough horizontal surface by a rope that is at an angle of 20° to the horizontal. The coefficient of friction between the block and the surface is 0.4.

(a) The tension in the rope is 150 newtons.

(i) Draw a diagram to show the forces acting on the block as it moves.

(2)

(ii) Show that the magnitude of the normal reaction force on the block is 243 newtons, correct to three significant figures.

(3)

(iii) Find the magnitude of the friction force acting on the block.

(2)

(iv) Find the acceleration of the block.

(4)

(b) When the block is moving, the tension is reduced so that the block moves at a constant speed, with the angle between the rope and the horizontal unchanged. Find the tension in the rope when the block is moving at this constant speed.

(5)

(c) If the block were made to move at a greater **constant** speed, again with the angle between the rope and the horizontal unchanged, how would the tension in this case compare to the tension found in part (b)?

(1)

(Total 17 marks)

Q18.

A particle, of mass 3 kg, moves along a straight line. At time t seconds, the displacement, s metres, of the particle from the origin is given by $s = 8t^3 + 15$

(a) Find the velocity of the particle at time t.

(2)

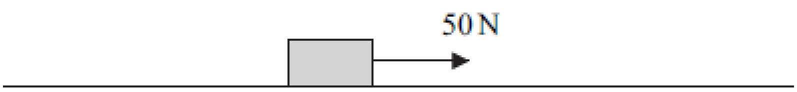
(b) Find the magnitude of the resultant force acting on the particle when $t = 2$.

(4)

(Total 6 marks)

Q19.

A block, of mass 4 kg, is made to move in a straight line on a rough horizontal surface by a horizontal force of 50 newtons, as shown in the diagram.



Assume that there is no air resistance acting on the block.

(a) Draw a diagram to show all the forces acting on the block.

(1)

(b) Find the magnitude of the normal reaction force acting on the block.

(1)

(c) The acceleration of the block is 3 m s^{-2}. Find the magnitude of the friction force acting on the block.

(3)

(d) Find the coefficient of friction between the block and the surface.

(2)

(e) Explain how and why your answer to part (d) would change if you assumed that air resistance did act on the block.

(2)

(Total 9 marks)

Q20.

A car, of mass 1200 kg, tows a caravan, of mass 1000 kg, along a straight horizontal road. The caravan is attached to the car by a horizontal towbar. A resistance force of magnitude R newtons acts on the car and a resistance force of magnitude $2R$ newtons acts on the caravan. The car and caravan accelerate at a constant 1.6 m s^{-2} when a driving force of magnitude 4720 newtons acts on the car.

(a) Find R.

(4)

(b) Find the tension in the towbar.

(3)

(Total 7 marks)

Q21.

A cyclist freewheels, with a constant acceleration, in a straight line down a slope. As the cyclist moves 50 metres, his speed increases from 4 m s^{-1} to 10 m s^{-1}.

(a) (i) Find the acceleration of the cyclist.

(3)

(ii) Find the time that it takes the cyclist to travel this distance.

(3)

(b) The cyclist has a mass of 70 kg. Calculate the magnitude of the resultant force acting on the cyclist.

(2)

(c) The slope is inclined at an angle α to the horizontal.

(i) Find α if it is assumed that there is no resistance force acting on the cyclist.

(3)

(ii) Find α if it is assumed that there is a constant resistance force of magnitude 30 newtons acting on the cyclist.

(3)

(d) Make a criticism of the assumption described in part (c)(ii).

(1)
(Total 15 marks)

Q22.

A particle, of mass 50 kg, moves on a smooth horizontal plane. A single horizontal force

$$[(300t - 60t^2)\mathbf{i} + 100e^{-2t}\,\mathbf{j}] \text{ newtons}$$

acts on the particle at time t seconds.

The vectors **i** and **j** are perpendicular unit vectors.

(a) Find the acceleration of the particle at time t.

(2)

(b) When $t = 0$, the velocity of the particle is $(7\mathbf{i} - 4\mathbf{j})$ m s^{-1}.

Find the velocity of the particle at time t.

(4)

(c) Calculate the speed of the particle when $t = 1$.

(4)
(Total 10 marks)

Q23.

A car is travelling at a speed of 20 m s^{-1} along a straight horizontal road. The driver applies the brakes and a constant braking force acts on the car until it comes to rest. Assume that no other horizontal forces act on the car.

(a) After the car has travelled 75 metres, its speed has reduced to 10 m s^{-1}. Find the acceleration of the car.

(3)

(b) Find the time taken for the speed of the car to reduce from 20 m s^{-1} to zero.

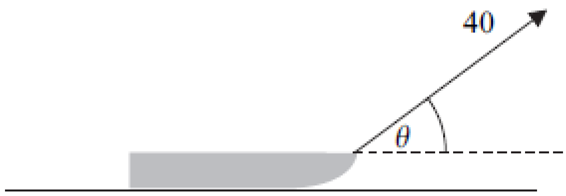
(2)

(c) Given that the mass of the car is 1400 kg, find the magnitude of the constant braking force.

(2)
(Total 7 marks)

Q24.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is 40 newtons. The rope is kept at an angle of θ to the horizontal, as shown in the diagram.



Model the sledge as a particle.

(a) Draw a diagram to show all the forces acting on the sledge.

(1)

(b) Find the magnitude of the normal reaction force acting on the sledge, in terms of θ.

(3)

(c) The acceleration of the sledge is

$$p + q\cos\theta + r\sin\theta$$

Find p, q and r.

(5)

(Total 9 marks)

Q25.

A car is travelling at a speed of 20 m s^{-1} along a straight horizontal road. The driver applies the brakes and a constant braking force acts on the car until it comes to rest.

(a) Assume that no other horizontal forces act on the car.

(i) After the car has travelled 75 metres, its speed has reduced to 10 m s^{-1}. Find the acceleration of the car.

(3)

(ii) Find the time taken for the speed of the car to reduce from 20 m s^{-1} to zero.

(2)

(iii) Given that the mass of the car is 1400 kg, find the magnitude of the constant braking force.

(2)

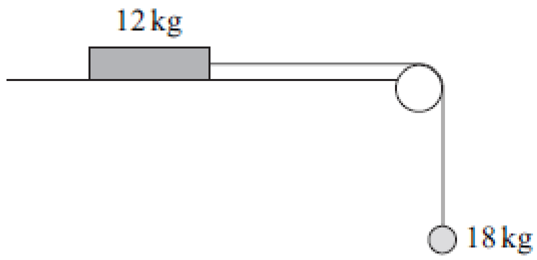
(b) Given that a constant air resistance force of magnitude 200 N acts on the car during the motion, find the magnitude of the constant braking force.

(1)

(Total 8 marks)

Q26.

A block, of mass 12 kg, lies on a horizontal surface. The block is attached to a particle, of mass 18 kg, by a light inextensible string which passes over a smooth fixed peg. Initially, the block is held at rest so that the string supports the particle, as shown in the diagram.



The block is then released.

(a) Assuming that the surface is smooth, use two equations of motion to find the magnitude of the acceleration of the block and particle.

(4)

(b) In reality, the surface is rough and the acceleration of the block is 3 m s^{-2}.

(i) Find the tension in the string.

(3)

(ii) Calculate the magnitude of the normal reaction force acting on the block.

(1)

(iii) Find the coefficient of friction between the block and the surface.

(5)

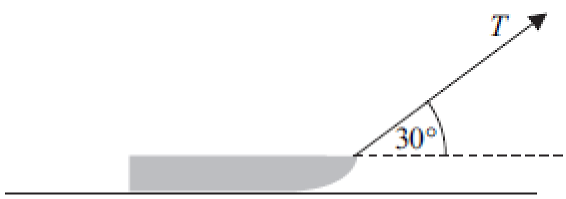
(c) State two modelling assumptions, other than those given, that you have made in answering this question.

(2)

(Total 15 marks)

Q27.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is T newtons. The rope is kept at an angle of 30° to the horizontal, as shown in the diagram.



Model the sledge as a particle.

(a) Draw a diagram to show all the forces acting on the sledge.

(1)

(b) Find the magnitude of the normal reaction force acting on the sledge, in terms of T.

(3)

(c) Given that the sledge accelerates at 0.05 m s^{-2}, find T.

(6)
(Total 10 marks)

Q28.

A particle moves in a straight line. At time t seconds, it has velocity v m s^{-1}, where

$$v = 6t^2 - 2e^{-4t} + 8$$

and $t \geq 0$.

(a) (i) Find an expression for the acceleration of the particle at time t.

(2)

(ii) Find the acceleration of the particle when $t = 0.5$.

(2)

(b) The particle has mass 4 kg.

Find the magnitude of the force acting on the particle when $t = 0.5$.

(1)

(c) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t.

(4)
(Total 9 marks)

Q29.

A stone, of mass 5 kg, is projected vertically downwards, in a viscous liquid, with an initial speed of 7 m s^{-1}.

At time t seconds after it is projected, the stone has speed v m s^{-1} and it experiences a resistance force of magnitude $9.8v$ newtons.

(a) When $t \geq 0$, show that

$$\frac{dv}{dt} = -1.96(v - 5)$$

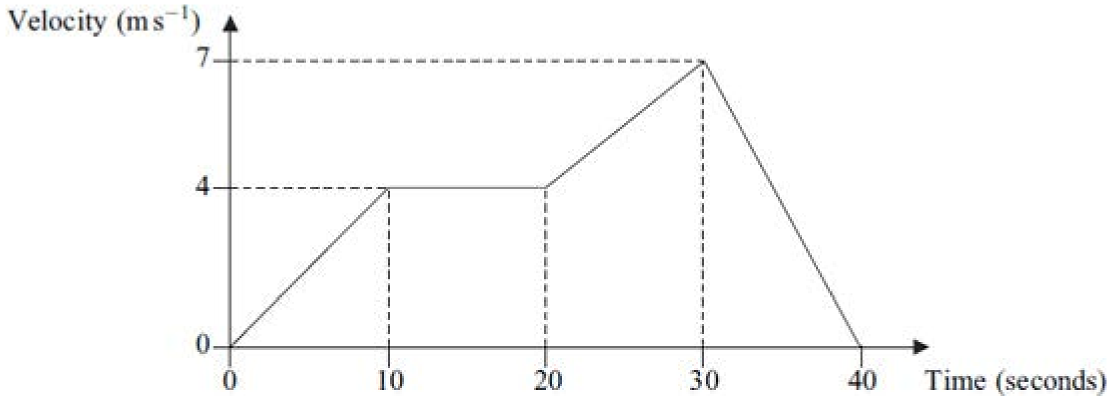
(2)

(b) Find v in terms of t.

(5)
(Total 7 marks)

Q30.

The graph shows how the velocity of a train varies as it moves along a straight railway line.



(a) Find the total distance travelled by the train.

(4)

(b) Find the average speed of the train.

(2)

(c) Find the acceleration of the train during the first 10 seconds of its motion.

(2)

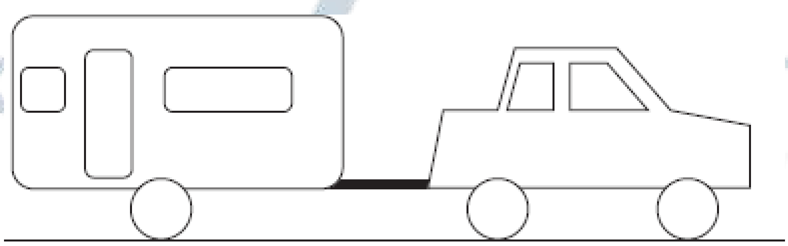
(d) The mass of the train is 200 tonnes. Find the magnitude of the resultant force acting on the train during the first 10 seconds of its motion.

(2)

(Total 10 marks)

Q31.

A car, of mass 1200 kg, tows a caravan, of mass 1000 kg, along a straight horizontal road. The caravan is attached to the car by a horizontal tow bar, as shown in the diagram.



Assume that a constant resistance force of magnitude 200 newtons acts on the car and a constant resistance force of magnitude 300 newtons acts on the caravan.
A constant driving force of magnitude *P* newtons acts on the car in the direction of motion.
The car and caravan accelerate at 0.8 m s^{-2}.

(a) (i) Find *P*.

(3)

(ii) Find the magnitude of the force in the tow bar that connects the car to the caravan.

(3)

(b) (i) Find the time that it takes for the speed of the car and caravan to increase from 7 m s^{-1} to 15 m s^{-1}.

(3)

(ii) Find the distance that they travel in this time.

(3)

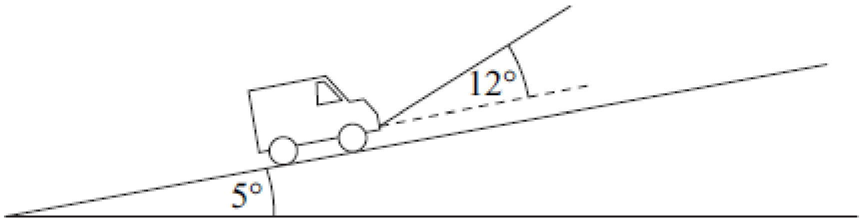
(c) Explain why the assumption that the resistance forces are constant is unrealistic.

(1)

(Total 13 marks)

Q32.

A van, of mass 2000 kg, is towed up a slope inclined at 5° to the horizontal. The tow rope is at an angle of 12° to the slope. The motion of the van is opposed by a resistance force of magnitude 500 newtons. The van is accelerating up the slope at 0.6 m s^{-2}.



Model the van as a particle.

(a) Draw a diagram to show the forces acting on the van.

(2)

(b) Show that the tension in the tow rope is 3480 newtons, correct to three significant figures.

(5)

(Total 7 marks)

Q33.

The velocity of a particle at time t seconds is $\mathbf{v}$ m s^{-1}, where

$$\mathbf{v} = (4 + 3t^2)\mathbf{i} + (12 - 8t)\mathbf{j}$$

(a) When $t = 0$, the particle is at the point with position vector $(5\mathbf{i} - 7\mathbf{j})$ m.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t.

(4)

(b) Find the acceleration of the particle at time t.

(2)

(c) The particle has mass 2 kg.

Find the magnitude of the force acting on the particle when $t = 1$.

(4)

(Total 10 marks)

Q34.

Vicky has mass 65 kg and is skydiving. She steps out of a helicopter and falls vertically. She then waits a short period of time before opening her parachute. The parachute opens at time $t = 0$ when her speed is 19.6 m s^{-1}, and she then experiences an air resistance force of magnitude $260v$ newtons, where v m s^{-1} is her speed at time t seconds.

(a) When $t > 0$:

(i) show that the resultant downward force acting on Vicky is

$$65(9.8 - 4v) \text{ newtons}$$

(1)

(ii) show that $\frac{dv}{dt} = -4(v - 2.45)$.

(2)

(b) By showing that $\int \frac{1}{v - 2.45} dv = -\int 4dt$, find v in terms of t.

(5)

(Total 8 marks)

Q35.

A crane is used to lift a load, using a single vertical cable which is attached to the load. The load accelerates uniformly from rest. When it has risen 0.9 metres, its speed is 0.6 m s^{-1}.

(a) (i) Show that the acceleration of the load is 0.2 m s^{-2}.

(3)

(ii) Find the time taken for the load to rise 0.9 metres.

(2)

(b) Given that the mass of the load is 800 kg, find the tension in the cable while the load is accelerating.

(3)

(Total 8 marks)